

CS505 Project Status Report: Citation-Grounded QA with Local Models

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1 Project Goal

The project studies hallucination reduction in scientific question answering using a fully local retrieval-augmented generation pipeline. The current stack is built around QASPER, shared SCC storage, and locally-run small instruction-tuned models. The main experimental question is how much answer quality and hallucination behavior depend on retrieval quality versus generation behavior, especially under citation forcing.

2 SCC Layout and Reproducibility

The implementation uses a strict split between shared deterministic artifacts and per-user experimental outputs.

Shared course path

```
/projectnb/cs505am/projects/zukhriddin_nlp_research_project/qasper
```

Student working repo

```
/projectnb/cs505am/students/zfmsai/zukhriddin_nlp_research_project
```

Backward-compatible symlink

```
/projectnb/cs505am/students/zfmsai/nlp_research_project
```

The code automatically resolves either the renamed project path or the older path. Shared data and processed artifacts live in `projects/`; model caches, runs, and temporary indexes live in `students/$USER/`.

3 What Was Implemented

3.1 Task 2: Data and Chunking

Implemented components:

- QASPER loader from official JSON/Hugging Face format
- Internal normalized schema for papers, questions, answers, and chunks
- Fixed chunking
- Sliding-window chunking
- Section-aware chunking
- Deterministic processed split builder for team-shared reuse

Canonical shared processed artifacts were built for:

- `train`
- `validation`
- `test`

These outputs are the interface boundary used by both retrieval and generation.

3.2 Task 3: Retrieval

Implemented retrieval stack:

- BM25
- Dense retrieval using `BAAI/bge-small-en-v1.5`
- Hybrid reciprocal-rank fusion
- Cross-encoder reranking using `cross-encoder/ms-marco-MiniLM-L-6-v2`
- Evaluation via Recall@5, MRR@10, and evidence hit rate

3.3 Task 4: Generation and Evaluation

Implemented generation stack:

- Baseline prompt style
- Citation-forcing prompt style
- Local generation with `microsoft/Phi-3.5-mini-instruct` via `transformers`
- Answer normalization for canonical QASPER answer forms (yes, no, unanswerable)
- Context compression to question-relevant sentences
- SCC-safe batch scripts and memory settings
- Evaluation via token F1, citation precision, citation rate, hallucination proxy, and supported-answer rate

3.4 Task 1: Infra and SCC

Implemented infrastructure:

- Reproducible path/config system
- SCC batch scripts for retrieval and generation
- Student-local Hugging Face cache handling
- Documentation for SCC environment setup
- GPU-safe defaults that run on `academic-gpu`

4 How the Pipeline Works

1. Load QASPER and normalize into internal objects.
2. Build processed split artifacts shared across the team.
3. Chunk each paper with one of the chunking strategies.
4. Retrieve from chunks belonging only to the paper for the given question.
5. Generate an answer from the top retrieved chunks.
6. Score the generated answer against gold answers and gold evidence.

The current best retriever for generation is `hybrid`; `hybrid_rerank` was evaluated later as a stronger comparator.

5 Main Results

5.1 Retrieval Results on Validation (Section Chunks)

Method	Recall@5	MRR@10	Evidence Hit
BM25	0.5769	0.4110	0.7152
Dense	0.5886	0.4820	0.7238
Hybrid	0.6313	0.5097	0.7778
Hybrid + rerank	0.6410	0.5027	0.7702

Table 1: Validation retrieval results on section-aware chunks.

Takeaway: Hybrid retrieval is the strongest overall default for generation; hybrid + rerank improves recall slightly but did not clearly dominate all retrieval metrics.

5.2 Generation Results with Hybrid Retrieval

Prompt	Questions	Token F1	Citation Prec.	Halluc.	Supported
Baseline	1005	0.3187	0.0000	0.4657	0.7612
Citation forcing	1005	0.3167	0.3423	0.5572	0.5741

Table 2: Full-validation generation results using hybrid retrieval. Citation forcing had citation rate 0.9254.

Takeaway: Citation forcing now works much better than the initial versions and produces citations reliably, but baseline remains better on answer quality, hallucination proxy, and supported-answer rate.

5.3 Generation Results with Hybrid + Rerank

Prompt	Questions	Token F1	Citation Prec.	Halluc.	Supported
Baseline	100	0.3697	0.0000	0.4300	0.8000
Citation forcing	100	0.3569	0.3528	0.5400	0.6600

Table 3: 100-question validation sample using hybrid + rerank. Citation forcing had citation rate 0.9100.

Takeaway: The strongest generation setting observed so far is the `hybrid_rerank` baseline. Even under stronger retrieval, citation forcing still trades off answer quality for citation behavior.

6 What Changed During Debugging

Several implementation issues were identified and fixed during the experiments:

- SCC scheduler instability prevented job submission for a period; jobs were moved to proper batch usage once SCC recovered.
- The correct GPU queue for this course environment was `academic-gpu`.
- Generation initially hit CUDA OOM; the fix was to cap input/context lengths and use conservative CUDA settings.
- Token F1 was artificially low because the model often produced verbose forms like “The context does not specify...” instead of exact canonical answers. Answer normalization fixed this.
- SCC environment-variable submission did not reliably propagate `QUESTION_LIMIT`; the batch script was changed to accept it as a positional argument.
- After the SCC project rename, shared path resolution failed; config detection was extended to support both old and new project names.

7 Current Status

Completed:

- Task 2: complete
- Task 3: complete
- Task 4 core Phi-based evaluation pipeline: complete
- Task 1 setup, SCC usage, and reproducibility documentation: complete for the current stack

Reached goals:

- deterministic shared QASPER preprocessing for the whole team
- full retrieval implementation and benchmarking
- full generation implementation and benchmarking with Phi-3.5
- SCC-compatible batch execution for both retrieval and generation
- clear best current setting before moving on to new models

Best current setting before Qwen:

- retrieval: hybrid_rerank is the strongest generation-side comparator on the current 100-question sample
- generation: baseline prompting is still better than citation forcing for raw answer quality

8 How to Inspect Results

From SCC:

```
cd /projectnb/cs505am/students/$USER/zukhriddin_nlp_research_project
ls -lt runs/retrieval/validation/section/*.json | head
ls -lt runs/generation/validation/section/*.json | head
```

To print metrics from a saved run:

```
python - <<'PY'
import json
from pathlib import Path
run_dir = Path("runs/generation/validation/section")
path = run_dir / "hybrid.baseline.20260409T011901Z.json"
obj = json.loads(path.read_text())
print(obj["metrics"])
PY
```

9 How to Reproduce the Pipeline

9.1 Environment

```
chmod +x setup_env.sh
./setup_env.sh
```

9.2 Check resolved paths

```
python3 scripts/show_project_paths.py
```

9.3 Retrieval evaluation

```
python scripts/evaluate_gaspar_retrieval.py \
    --split validation \
    --strategy section \
    --method hybrid
```

9.4 Generation evaluation

```
python scripts/evaluate_gaspar_generation.py \
    --split validation \
    --strategy section \
    --retrieval-method hybrid \
    --prompt-style baseline \
    --generation-model microsoft/Phi-3.5-mini-instruct
```

9.5 SCC batch generation sample

```
qsub -N gaspar_gen_sample -o logs/gaspar_gen_sample.log \
    scripts/qsub_gaspar_generation.sh hybrid citation_forcing \
    validation section 5 20 100 2048 1200 128
```

The final four positional arguments are:

- question limit
- max input tokens
- max context tokens

- max new tokens

10 Next Step

The current Phi-based milestone is complete enough to move on. The next experiments should compare:

- Qwen2.5-7B-Instruct
- Qwen3-4B-Instruct-2507

Those runs should reuse the same retrieval and evaluation pipeline so that model changes are isolated from data and retrieval changes.